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PDX Group 3

Clinical Case on Sensitivity and Specificity

Background:

A 60-year-old man visits his primary care physician for a routine checkup. He has been feeling well overall but has a history of smoking, and his physician is concerned about the risk of chronic obstructive pulmonary disease (COPD). As part of his routine screening, the physician recommends a **spirometry test**, which measures lung function and helps diagnose COPD.

The Test:

The spirometry test categorizes patients based on their lung function:

1. **Positive result**: The test suggests that the patient has COPD or significant airflow limitation.
2. **Negative result**: The test suggests normal lung function without COPD.

For this test, the following characteristics are given:

- **Sensitivity**: 80% (The test correctly identifies 80% of patients who have COPD).
- **Specificity**: 85% (The test correctly identifies 85% of patients who do not have COPD).

Clinical Scenario:

The physician explains that the accuracy of the test can be assessed using the metrics of sensitivity and specificity. The prevalence of COPD in this patient's age group (60-year-old men with a history of smoking) is around 20%.

Case Analysis:

1. **True Positive (TP)**: If the patient has COPD, the test will correctly identify it 80% of the time.
2. **False Negative (FN)**: If the patient has COPD, there is a 20% chance the test will miss it (a false negative).
3. **True Negative (TN)**: If the patient does not have COPD, the test will correctly identify this 85% of the time.
4. **False Positive (FP)**: If the patient does not have COPD, there is a 15% chance the test will incorrectly suggest that he does (a false positive).

Example Calculations:

Out of 1000 men who are screened (assuming 20% prevalence of COPD):

Given:

- Total population = 1,000 men
- Prevalence = 20%
- Sensitivity = 80%
- Specificity = 85%

So, With COPD: Population x Prevalence = 1,000 x 0.20 = **200**
And, Without COPD: Population – w/ COPD = 1,000 – 200 = **800**

Calculate:

- **True Positives**: 200 x 0.80 = **160**
- **False Negatives**: 200 x 0.20 = **40**
- **True Negatives**: 800 x 0.85 = **680**
- **False Positives**: 800 x 0.15 = **120**

Implications:

- **Sensitivity** of 80% means the test correctly identifies 80% of individuals who have COPD, while missing 20% of the cases.
- **Specificity** of 85% means the test correctly identifies 85% of individuals who do not have COPD, while incorrectly classifying 15% as positive.
- **False Positives**: means individuals who were detected as having a positive test result but didn't have COPD.
- **False Negatives**: means individuals who were detected as having a negative test result but did have COPD.

Conclusion:

The spirometry test has a sensitivity of about 80% and specificity of 85%, indicating that it identifies most individuals correctly with COPD, and correctly classify most individuals as well with no COPD. In **1,000 men**, **160** would be **correctly classified** as **having COPD**, **missing 40** cases in tandem and **680** individuals **without COPD would be properly identified** while **120 would receive false positive** results. Thus, spirometry is a good screening test and would be useful for identifying possible COPD cases. However, it should be noted that we should still interpret positive and negative results in consideration of the patient's clinical condition of other diagnostic findings.